Databases for Big Data

Exercises: Data Formats & Encoding

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Exercises

Exercise 1.

The following schema of a structure is given.

```
message StudentRecord {
  required int32 StudentID;
  required string Name;
  optional int32 Age;
  optional group Address {
    required string City;
    optional string ZipCode;
  }
  optional group Enrollment {
    required string Program;
  }
}
```

An instance of the above schema is as follows.

```
{
  "StudentID": 1001,
  "Name": "Emma Liu",
  "Age": 21,
  "Address": {
    "City": "Toronto",
    "ZipCode": "M5V 2T6"
  },
  "Enrollment": {
    "Program": "Computer Science"
  }
}
```

Convert the above record in columns as defined in Parquet format.

Exercise 2.

Convert the record presented in Exercise 1 in columns as defined by ORC format.

Descriptive and explanatory questions.

1. Present the storage models for OLAP DBMS that define the physical organization of tuples on disk and in memory.

2. Describe the PAX data format and its descendant, the Parquet data format.

3. Describe the metadata included in a Parquet file. Which metadata is missing that is needed in parallel and distributed OLAP DBMSs?

4. Present the optimizations that Parquet uses to speed up data access and processing.

5. List the data encoding schemes used by the OLAP DBMSs. Describe briefly each encoding.

6. Which encodings would you use for compressing HTML files?

Clarification: Record shredding in Dremel

From: Melnik, et.al., Dremel: Interactive Analysis of Web-Scale Datasets, VLDB, 2010.

r_1
```
DocId: 10
Links
  Forward: 20
  Forward: 40
  Forward: 60
Name
  Language
    Code: 'en-us'
    Country: 'us'
  Language
    Code: 'en'
  Url: 'http://A'
Name
  Url: 'http://B'
Name
  Language
    Code: 'en-gb'
    Country: 'gb'
```

```
message Document {
  required int64 DocId;
  optional group Links {
    repeated int64 Backward;
    repeated int64 Forward; }
  repeated group Name {
    repeated group Language {
      required string Code;
      optional string Country; }
    optional string Url; }}
```

r_2
```
DocId: 20
Links
  Backward: 10
  Backward: 30
  Forward:  80
Name
  Url: 'http://C'
```

DocId

value	r	d
10	0	0
20	0	0

Name.Url

value	r	d
http://A	0	2
http://B	1	2
NULL	1	1
http://C	0	2

Links.Forward

value	r	d
20	0	2
40	1	2
60	1	2
80	0	2

Links.Backward

value	r	d
NULL	0	1
10	0	2
30	1	2

Name.Language.Code

value	r	d
en-us	0	2
en	2	2
NULL	1	1
en-gb	1	2
NULL	0	1

Name.Language.Country

value	r	d
us	0	3
NULL	2	2
NULL	1	1
gb	1	3
NULL	0	1

1. *Repetition levels:*
   Tells us at what repeated field in the field's path the value has repeated.

2. *Definition levels*:
   Each value of a field with path p, especially, every NULL, has a definition level specifying how many fields in p that could be undefined (because they are optional or repeated) are actually present in the record.